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Abstract

This project presents an autonomous, closed-loop reinforcement learning (RL) control system for resonator frequency calibration, motivated by the calibration bottleneck in superconducting quantum computing. A classical LC resonator circuit serves as a hardware analogue for a superconducting qubit, and a Proximal Policy Optimization (PPO) agent is trained to maintain peak amplitude tracking under stochastic environmental drift. Four simulation models (V1–V4) were developed iteratively, with the final V4 architecture introducing a normalised amplitude gradient term into the observation space. V4 achieved a 34% reduction in mean absolute tracking error (MAE) compared to the V3 baseline under Gaussian noise, with statistical significance $p < 0.001$. The trained policy was further deployed to a simulated quantum circuit using Qiskit Aer (QACT pipeline), confirming sim-to-real transferability. Hardware implementation using ESP32, MCP4921 DAC, MCP3202 ADC, and a varactor-tuned LC tank circuit is in progress.

Introduction and Motivation

Problem

Superconducting qubits are extremely sensitive to stochastic magnetic flux noise, which causes continuous drift in their resonant frequency over time. At present, recalibration is mainly done using brute-force frequency sweeps or PID controllers. However, both methods are slow, computationally inefficient, and struggle to work properly on non-linear Lorentzian response curves of resonator systems.

Classical Analogue

Instead of directly working with expensive cryogenic quantum hardware, this project uses a classical LC resonator operating around 500 kHz as a practical analogue for a superconducting qubit. The reason for this is that both systems follow the same Lorentzian response equation:

$$A(f) = 1 / [1 + Q^2((f-f_0)/f_0)^2]$$

Because of this shared mathematical behaviour, an RL policy trained on a classical resonator can potentially transfer its control strategy into the quantum domain as well.

Approach

In this work, a PPO based reinforcement learning agent is trained in simulation to continuously track the drifting resonant peak using only scalar amplitude measurements. The main contribution of the final V4 model is the addition of an explicit normalised gradient signal into the observation space. This allows the agent to distinguish whether an amplitude drop is caused by its own movement or by environmental drift, effectively solving the gradient blindness problem that affects both traditional PID controllers and earlier RL models.

System Architecture and Reinforcement Learning Framework

The proposed system consists of both hardware and software layers working together in a closed-loop control setup. The hardware signal chain was designed using five major stages – a regulated power supply, ESP32 with MCP4921 DAC for waveform generation, an LC resonator circuit, an envelope detector, and an MCP3202 ADC for amplitude readback. The LC resonator acts as the classical analogue for a superconducting qubit and was tuned close to 556 kHz using a 10 μ H inductor and varactor based capacitor network. LTSpice simulations confirmed a sharp Lorentzian resonance curve with high Q-factor behaviour. The software stack separates the resonator physics simulation, RL environment, training pipeline, and evaluation modules. Different Python scripts were developed for stochastic drift simulation, PPO based environment training, and long episode evaluation using metrics such as MAE and near-resonance tracking percentage.

The reinforcement learning framework was developed iteratively from V1 to V4 to overcome different failure modes observed during training. Early models suffered from problems such as static lock and overshoot oscillations, while V3 achieved stable tracking but still struggled under extreme stochastic drift because of gradient blindness. The final V4 model introduced an additional parameter called `amp_gradient` into the observation space, which acts as a finite difference estimate of the Lorentzian slope. This helped the agent distinguish between amplitude drops caused by its own movement and those caused by environmental drift. The PPO agent used a reward function that maximised resonance proximity while penalising unstable frequency jumps. The model was tested under both Gaussian random walk noise and Ornstein-Uhlenbeck noise conditions to evaluate robustness. Results showed that the gradient augmented V4 model achieved significantly lower tracking error and more stable performance under difficult non-stationary drift environments.

Results and Statistical Analysis

Table 1 – Performance Comparison Under Gaussian Noise

Controller	MAE (Hz)	Amp > 0.90 (%)	Amp < 0.70 (%)
PID	3600 ± 5241	83.1% ± 18.5%	5.8% ± 19.1%
V3 (RL)	3333 ± 1932	64.50%	13.40%
V4 (RL + Gradient)	2205 ± 518	84.8% ± 7.6%	5.1% ± 3.7%

Table 2 – Performance Comparison Under OU Noise

Controller	MAE (Hz)	Amp > 0.90 (%)	Amp < 0.70 (%)
PID	1414 ± 262	95.8% ± 5.1%	0.0% ± 0.1%
V3 (RL)	1241 ± 254	97.70%	0.00%
V4 (RL + Gradient)	1284 ± 93	97.9% ± 0.4%	0.00%

Table 3 – Statistical Significance Analysis

Noise Regime	Comparison	p-value	Verdict
Gaussian Noise	V3 vs V4	p < 0.001	Significant
OU Noise	V3 vs V4	p = 1.000	Not Significant

Under Gaussian drift conditions, the gradient augmented V4 model achieved the best overall performance with nearly 34% lower MAE compared to the V3 baseline model. The standard deviation also reduced significantly, showing that V4 produced more stable and reliable tracking behaviour. In comparison, PID controllers showed very unstable performance under strong stochastic drift. Under OU noise conditions however, all three controllers performed similarly because the mean reverting drift rarely pushed the resonator far enough away from resonance to expose gradient blindness effects.

Qiskit Aer Quantum Circuit Simulation (QACT)

The QACT (Quantum Agent Control Test) pipeline was developed to test whether the PPO agent trained on the classical LC resonator could transfer its control strategy into a simulated quantum environment using Qiskit Aer. In this setup, the agent controls a two-level qubit whose excitation probability follows a Lorentzian response similar to the classical resonator system. Different QACT versions were tested to improve transfer performance step by step. The Zero-Shot model directly used simulation trained weights without fine tuning, while later versions introduced fixes such as environment compatibility updates and VecNormalize corrections. Among all versions, QACT v4 achieved the best MAE performance of nearly 2200 Hz, while QACT v5 achieved the highest near-resonance percentage of about 34%. Although the quantum simulation performance still remained below the original simulation target of 94.8% near-resonance, the results confirmed partial sim-to-real transfer capability. The remaining performance gap is mainly due to quantum measurement noise and the absence of hardware-in-the-loop fine tuning, which is planned in the next phase of the project.

Table 4 – QACT Version Performance Comparison

Version	Key Change	MAE (Hz)	Near-Resonance (%)
Zero-Shot	V4 Gaussian weights without fine-tuning	4400 ± 1100	22%
QACT v2	Fine-tuned with incorrect learning rate	4100 ± 1000	32%
QACT v3	Environment compatibility fix	$3300 \pm \text{—}$	32%
QACT v4	VecNormalize fix applied	2200 ± 900	27%
QACT v5 (Seed 3)	Best simulation generalisation weights	3200 ± 4581	34%

Remaining Work and Future Roadmap

Immediate Simulation Tasks

- Simulate MCP4921 DAC SPI communication and MCP3202 ADC readback using Proteus 8.5.
- Perform V4a and V4b ablation studies to isolate the contribution of gradient normalisation.
- Conduct robustness testing for V3 and V4 Seed 3 models at higher unseen noise levels ($\sigma = 750$ Hz and 1000 Hz).
- Extract TensorBoard logs and generate sample efficiency curves for near-resonance convergence analysis.

Hardware Implementation

- Assemble the complete five-stage signal chain on breadboard hardware.
- Perform frequency sweep experiments to measure practical resonant frequency and Q-factor values.
- Deploy the V4 Seed 3 PPO model onto ESP32 hardware using Hardware-in-the-Loop (HIL) fine tuning.
- Optimise the controller for varactor non-linearities and serial communication delays.

Paper Writing

- Prepare complete research documentation including methodology, results, and related work.
- Organise experimental plots, trajectory analysis, and statistical evaluation results.
- Target submission towards undergraduate IEEE conferences and arXiv preprint publication.

Conclusion

This project demonstrates that a reinforcement learning based control system with gradient augmentation can outperform both traditional PID controllers and earlier gradient blind RL models for autonomous resonator frequency tracking under stochastic drift conditions. By adding the `amp_gradient` parameter into the observation space, the V4 model achieved a statistically significant 34% reduction in MAE along with much more stable tracking performance. The trained policy was also successfully tested on a Qiskit Aer quantum simulation, showing partial sim-to-real transfer capability and validating the Lorentzian based control approach. Future work mainly focuses on full ESP32 hardware deployment and Hardware-in-the-Loop fine tuning for real-time resonator calibration.